Software Requirements Specification

SHIELD TECHNOLOGIES

Fruit Shop POS & Inventory Product

Version 2.0 | 14 October 2026 | Draft for approval | Subscription-based SaaS

1. Purpose

Shield Technologies will provide a subscription-based web POS and inventory product for fruit retailers. It shall support manual data entry, retail checkout, purchasing and receiving, batch inventory, ripeness tracking, spoilage control, stock counts, and management reports. This specification replaces the generic version 1.0 scope with a fruit shop scope.

2. Scope and assumptions

The first release supports one shop per subscribing business, with storage and display locations inside that shop. It supports kilogram-based fruit and count-based items, configurable tax, LKR as the default currency, and Asia/Colombo as the default timezone. Each business may configure its own currency and timezone. Requirements use “shall” for mandatory behavior.

Staff shall enter master data, weights, counts, receipts, ripeness assessments, wastage, and stock counts manually. A scale reading is entered by staff; automatic scale, scanner, payment gateway, and accounting integrations are future enhancements. Cash and externally processed card payments are recorded; no card credentials are stored.

The product requires internet connectivity. Multi-branch inventory, offline synchronization, credit sales, supplier payment settlement, accounting ledgers, loyalty, delivery logistics, fruit processing, and mixed-fruit bundles are outside this release. Purchase orders and supplier returns are included; these were excluded in version 1.0.

3. Intended users

Shield administrator: tenant onboarding and subscription administration. Owner/manager: staff, prices, suppliers, purchasing, adjustments, refunds, and reports. Stock clerk: receiving, batch inspection, location movement, wastage requests, and stock counts. Cashier: checkout, receipts, product/stock lookup, and optional customer entry. One user may hold multiple roles.

4. Reading guide

Pages 2-5 define inventory data and rules. Pages 6-9 define functional requirements. Page 10 defines service boundaries. Pages 11-12 define non-functional requirements and acceptance. Pages 13-15 contain ER and UML use case diagrams. Page 16 defines operational exceptions and release decisions. Diagrams are logical models; service-owned references do not imply cross-service database joins.

5. Inventory: product and supplier master data

5.1 Product record - what must be stored

Required: tenant ID, immutable product ID, unique SKU within tenant, fruit/item name, category, variety, grade, base stock unit (KG or EACH), sale unit, selling price per base unit, tax rate, active status, low-stock threshold, reorder target, and whether batch tracking is required. All fruits shall be batch tracked. Optional: display code/PLU, description, origin, default supplier, storage guidance, reorder lead time, and indicative shelf-life days.

Example: Mango / Karthakolomban / Grade A / KG / LKR 650 per kg. A separate variety or grade shall be a separate SKU when it has a different selling price or stock pool. Renaming a product shall not change historical sale descriptions.

5.2 Units and conversions

Stock shall be maintained in one immutable base unit per SKU once transactions exist. KG quantities shall support three decimal places; EACH quantities shall be whole numbers. Grams shall convert to kilograms by dividing by 1,000. A price is per kg or per item, never per an unspecified unit.

A crate or carton is a receiving container, not a fixed stock unit. Staff shall enter its actual net fruit weight or item count. Supplier billing in crates may be recorded as container count and total invoice value; unit cost shall be total net line cost divided by accepted base quantity. Count-to-weight conversions shall not be inferred. A separately sold fixed pack must use its own SKU, with a controlled transfer if stock is repacked.

5.3 Category and supplier records

Category: ID, tenant, name, active status. Supplier: ID, tenant, supplier code, name, contact person, phone, optional email/address, active status, and notes. Deactivation shall preserve historical purchases and returns. The same supplier may provide multiple products and batches.

5.4 Price and reorder settings

Maintain current price, price history, effective time, actor, reason, discount limits, and optional batch clearance price. Owners shall change prices. Cashiers shall apply only permitted discounts. Products shall store reorder threshold and target separately: threshold triggers an alert; target supports a suggested order quantity.

Suggested order = max(0, target - sellable quantity - confirmed outstanding purchase quantity). This is advisory and requires owner review; seasonal demand forecasting is outside scope. All thresholds shall use the product base unit.

6. Inventory: batches and stock positions

6.1 Batch / lot record

Each received fruit line shall create a batch with batch ID, tenant ID, product ID, supplier ID, receipt line ID, received timestamp, supplier lot reference if available, accepted quantity, unit cost snapshot, origin if known, inspection notes, and optional harvest date. Multiple receipts shall create separate batches even when supplier lot references match.

Perishable attributes: ripeness (unripe, ready, overripe), expected sell-by date, optional use-by date, quality state (saleable, quarantined, spoiled), and assessment history. Staff-entered dates are estimates; the product does not certify food safety. Date changes shall require a reason and an audit entry.

6.2 Stock position

Stock shall be tracked by tenant, product, batch, and location. A stock position shall contain physical quantity, reserved quantity, and quality status. Available quantity = physical quantity - reserved quantity for saleable stock only. Quarantined, spoiled, or past-use-by stock shall have zero sellable availability. Reservations shall not reduce physical stock until checkout is committed.

Location records shall contain ID, code, name, and active status. Initial examples are Display and Store Room. Location transfers shall move quantity between positions in a single Inventory transaction without changing total shop stock. A location with positive stock shall not be deactivated.

6.3 Allocation rules

The default allocation shall use FEFO (first expiry / sell-by first), ordered by use-by if present, otherwise sell-by, then receipt date. Batches without dates shall follow dated batches and use FIFO among themselves. A sale may consume multiple batches. Allocation details shall be retained for traceability and cost reporting.

Ready fruit is the default sale pool. Owners may allow sale of unripe or overripe fruit if it is still classified saleable. Batch selection or clearance overrides require authorized permission and a reason. Past-sell-by stock shall require owner review and a recorded decision before sale; past-use-by stock shall be blocked until disposed of or returned, without overriding the original use-by date.

6.4 Opening stock

Owners shall enter opening quantity by product, batch, location, estimated cost, ripeness, and dates. When the supplier is unknown, opening stock may omit supplier reference and shall be clearly marked Opening Stock. An opening movement shall establish the initial balance; users shall never overwrite a balance field directly.

7. Inventory: receipt, loss and stock counting

7.1 Purchase orders and receiving

Purchase order header: supplier, order date, expected date, status, reference, notes. Lines: product, ordered base quantity, agreed unit cost, optional container count. Statuses: draft, approved, partially received, received, closed, cancelled. Only unreceived quantities may be cancelled. Excess receipt requires owner authorization; completed receipts shall not be edited.

Goods receipt header: supplier, linked order if any, delivery reference, received-by user, date, status. Lines: product, delivered quantity, accepted quantity, rejected quantity, gross weight, tare weight, net weight where relevant, actual line cost, batch data and destination location. Net weight = gross - tare, with gross >= tare. Accepted + rejected = delivered; quantities shall use the base unit.

Only accepted quantity shall enter stock. Rejected fruit remains outside owned stock and shall be recorded with reason. Receipts without an order shall be allowed to support market purchases. Corrections after posting shall use a reversal and replacement, blocked where later allocations would make reversal invalid.

7.2 Wastage and supplier returns

Wastage: batch, location, quantity, reason (spoilage, bruising, shrinkage, handling damage, theft, other), notes, reporter, approver, timestamp, and unit cost snapshot. Spoilage is a quality change followed by an authorized quantity removal. Shrinkage from dehydration is a separate measured loss; it shall not be hidden inside a sale.

Supplier return: supplier, original receipt/batch, location, quantity, reason, date, approval, and optional supplier credit reference. A return shall reduce physical stock exactly once. Supplier financial settlement is outside scope. Reserved quantity cannot be wasted, returned, or moved until its reservation is released or resolved.

7.3 Stock counts and adjustments

A count session shall store scope, batch/location positions, system snapshot, physical counted quantity, counted-by user, count time, variance, review notes, and approval. For the simple release, Inventory shall lock movements and checkout allocations affecting the scoped positions during counting. Users shall be warned; unrelated stock remains usable.

Variance = counted - snapshot. Only owner approval shall post a linked adjustment. Count approval and adjustment creation shall be atomic and idempotent. Cancellation shall unlock the scope without changing stock. Lost connection or an abandoned count shall require owner cancellation or recovery; counts shall never silently expire into adjustments.

8. Inventory: ledger, alerts and examples

8.1 Stock movement ledger

Every change shall create an immutable movement: tenant, movement ID, product, batch, location, signed base quantity, movement type, source document and line reference, reason, actor, approval, timestamp, and cost snapshot. Types: opening, accepted receipt, sale, sale return, transfer out/in, wastage, supplier return, count adjustment, and approved repacking transfer.

For each position: physical stock = sum of posted movement quantities. Transfers shall have linked equal-and-opposite movements. Posted movements shall be corrected by linked reversal documents, never deleted. Reservations shall be stored separately, with checkout reference, quantity, expiry time, and status.

8.2 Inventory screens and reports

Product list: SKU, fruit, variety/grade, price/unit, available quantity, reorder status. Batch list: supplier, receipt date, location, ripeness, dates, cost, physical, reserved and saleable quantity. Detail: complete movements, quality assessments and related receipts/sales/returns. Forms shall show unit labels beside every quantity and price.

Alerts shall identify low stock, zero sellable stock, near sell-by/use-by, overdue quality review, quarantined/spoiled stock awaiting disposal, and unresolved reservations. Owners shall configure the near-date window (default two calendar days). Alerts shall refresh at least every five minutes and on relevant stock changes.

Reports shall include stock by product/batch/location, ageing, near-date stock, supplier receipts, stock movement history, wastage quantity and value, count variance, and gross margin. Different units shall not be summed into one quantity total. Stock value = remaining batch quantity x batch unit cost; loss value = wasted quantity x original batch cost.

8.3 Worked examples / test fixtures

Receiving: 24.000 kg gross mangoes less 4.000 kg crates = 20.000 kg net. Accept 18.000 kg and reject 2.000 kg. Net accepted line cost LKR 7,200 produces LKR 400/kg batch cost. Stock increases by 18.000 kg only.

Checkout: sell 1.250 kg at LKR 650/kg = LKR 812.50 before discount/tax. A 10% line discount gives LKR 731.25 taxable amount. With tax configured at 0%, cash tender LKR 1,000 gives LKR 268.75 change. This is an example setting, not a statement of applicable tax law.

Remaining stock: 18.000 - 1.250 sale - 0.750 spoilage = 16.000 kg. Moving 5.000 kg to Display leaves total 16.000 kg. If a subsequent count finds 15.800 kg, an approved -0.200 kg adjustment records the shortage explicitly.

9. Functional requirements: access and product setup

FR-01 | Tenant onboarding

Shield administrators shall create tenant/store profiles, configure timezone/currency, and create the initial owner. Every business record and operation shall be tenant scoped.

FR-02 | Users and permissions

The system shall support login, password reset, role assignment, and account disablement. Owners shall manage only their own users. Cashiers shall not change costs, approve loss/returns, or edit master data. Stock clerks shall not change retail prices or approve their own adjustments.

FR-03 | Subscriptions

Shield administrators shall configure plans, record verified subscription payments, and activate, renew, suspend, or cancel subscriptions. Owners shall view status and renewal date. On expiry, business writes shall be blocked and owner read/export retained; already authorized pending transactions shall be reconciled. Cancellation stops renewal while retaining access through the paid period.

FR-04 | Fruit product setup

Owners shall maintain the product fields, varieties, grades, units, thresholds and price history defined in section 5. SKU shall be unique per tenant. Referenced products shall be deactivated, never deleted. Unit changes after stock movements shall require creation of a new SKU.

FR-05 | Supplier and location setup

Owners shall maintain suppliers, categories and shop locations. Clerk access shall be limited to viewing these records and selecting them on permitted documents. All referenced historical records shall remain available after deactivation.

FR-06 | Validation

The system shall reject missing required fields, negative prices, invalid dates, KG precision above three decimals, fractional EACH quantities, and invalid tare/quantity relations. Money shall use two decimal places; cost rates may store six decimal places. Zero-priced sales shall require owner permission and reason.

FR-07 | Price and tax control

Owners shall set tax-exclusive prices, tax rates and permitted discounts. Line amount = quantity x unit price; line discount reduces taxable value; line tax is added afterward. Monetary line amounts, discounts and taxes shall be rounded half-up to two decimals and totals summed from rounded lines. Receipts shall show the applied rates.

10. Functional requirements: inventory operations

FR-08 | Opening stock and purchase orders

Owners shall post opening stock and create/approve purchase orders. Orders shall support partial receipt, outstanding quantities, closure and cancellation as defined in section 7. Purchase suggestions shall remain advisory.

FR-09 | Receiving

Clerks shall draft and post approved-order receipts within ordered quantities; owners shall post direct or excess receipts. Receiving shall record acceptance/rejection, weights, cost, batch and location. Posting shall atomically create accepted stock and movements and update receipt status. Retry shall not duplicate stock.

FR-10 | Batch quality and date review

Clerks shall record ripeness and inspection results; owners shall approve release from quarantine and past-sell-by decisions. Quality/date changes shall retain before/after values and reason. Spoiled and past-use-by batches shall be excluded from sale allocations.

FR-11 | Availability, reservations and transfers

Inventory shall calculate available stock, reserve/commit/release quantities, enforce FEFO, and prevent overselling under concurrency. Clerks shall transfer unreserved stock between locations with linked movements; tenant totals shall stay unchanged.

FR-12 | Wastage and supplier returns

Clerks shall submit wastage and supplier return records. Only owner approval shall remove stock. The system shall retain reasons and original costs, validate remaining unreserved stock, and post each removal once. Cashiers shall not authorize these changes.

FR-13 | Stock counts

Clerks shall open a count scope, enter counts and submit variances. The system shall lock scoped positions against movements; owners shall approve adjustments or cancel the count. Audit shall show snapshot, count, variance and approval. Approval shall post one adjustment per position and unlock it.

FR-14 | Ledger and alerts

The system shall preserve the movement ledger and provide product/batch/location lookup and alerts defined in section 8. Historical ledger entries shall be immutable. Authorized corrections shall create linked reversals and shall not make stock negative.

11. Functional requirements: checkout and returns

FR-15 | Fruit checkout

Cashiers shall search by name/SKU/PLU, select product, enter kg or item quantity, see unit price and stock, and edit the basket before payment. For weighing, cashiers may enter gross and tare weights; the system shall calculate net sale quantity and reject zero/negative net weight. It shall not subtract packaging from stock.

FR-16 | Allocation and sale completion

Sales shall request batch reservations, retain allocation quantities and cost snapshots, and complete only after payment recording and Inventory commit are confirmed. Unavailable or locked stock shall block affected lines. An authorized batch override shall be audited. A repeated checkout identifier shall return the same transaction outcome.

FR-17 | Payment and receipt

The system shall record cash tender/change or externally confirmed card payment. Split payments are outside the MVP. Completed sales shall receive a unique number and printable A4/80 mm receipt with fruit, quantity/unit, price, discount, tax, total, cashier and timestamp. Reprinting shall not create another sale.

FR-18 | Failed checkout and voids

Users shall cancel an unpaid basket without stock change. Pending checkout shall show its status and reconcile before retry or cancellation. A completed sale shall not be deleted or voided through basket cancellation; corrections shall use a linked return/refund.

FR-19 | Sale returns and refunds

Owners shall return partial/full quantities against original sale lines. Cumulative quantity/value returned shall not exceed the original less earlier returns. Refunds shall use original price/discount/tax allocations, preserving final rounding balance. Refund payment shall be recorded as externally confirmed; the system does not transfer funds.

FR-20 | Returned fruit inspection

Returned fruit shall default to quarantine and shall not immediately increase sellable availability. Owners may release it to its original batch only when identity and quality are verified. Untraceable returns shall use a new quarantined return batch linked to the sale; spoiled returns shall be disposed with a wastage record. A refund without goods returned shall not add stock.

FR-21 | Optional customer record

Cashiers shall optionally capture customer name and contact number and attach them to a sale. Walk-in sales shall need no customer data. Access shall be tenant scoped; sensitive payment credentials shall never be entered.

12. Functional requirements: reports and administration

FR-22 | Sales and profitability reports

Owners shall view sales, refunds, net sales, payments and product sales by business date/range. Gross margin shall be net revenue excluding tax less batch cost of sold quantities, adjusted for returns. Wastage shall be shown separately. Reports shall show currency/unit, generation time and projection refresh time.

FR-23 | Inventory reporting and traceability

Owners shall access all section 8 inventory reports and export CSV. Batch traceability shall link receiving supplier to movements and sale allocations. Users shall search by batch, product, receipt or sale. Cost and margin data shall be hidden from cashiers.

FR-24 | Cashier shift summary

Cashiers shall open/close a shift with opening float, counted closing cash and reasoned variance. Expected cash = opening float + cash sale receipts - cash refunds + approved cash-in - approved cash-out. Owners shall approve cash movements and review shifts. No drawer hardware control is required.

FR-25 | Audit and settings

The system shall audit authentication failures, permission changes, prices, quality reviews, stock operations, sales/returns, subscription changes and exports. Owners shall configure business details, receipt text, timezone, thresholds and tax. Neither business users nor Shield business support screens shall edit posted audit records.

Document state conventions

Inventory documents: Draft > Submitted > Posted, or Rejected/Cancelled. Owner approval is mandatory for wastage, supplier returns and count adjustments; clerks cannot self-approve. Drafts have no stock effect. Posted records are immutable; reversal documents require owner authorization.

Sale: Draft > Pending > Completed or Failed/Cancelled. Sale return: Draft > Pending > Posted or Failed/Cancelled. Reservation: Active > Committed or Released/Expired. Each transition shall be recorded with its actor and time. Unknown outcomes shall stay Pending until reconciled.

Report reconciliation

Completed sales only shall contribute to sales reports. Posted returns shall reduce reported revenue. Physical stock shall reconcile with the movement ledger; reserved stock shall reconcile with active checkout reservations. Daily reconciliation shall identify discrepancies for an owner/support workflow, without silently altering data.

13. Microservices and consistency design

13.1 Service boundaries

Identity & Subscription: tenants, stores, users/roles, plans/subscriptions, subscription payments. Catalogue & Inventory: products/categories, suppliers, locations, purchases/receipts, batches/quality, positions/ reservations, movements, counts, wastage, supplier returns. Sales: sales/lines, allocations, retail payments, returns/refunds, customers, shifts/cash movements. Reporting: read-only projections and exports.

The browser shall use an HTTPS API gateway. Services shall own isolated schemas/stores and be independently deployable. Services shall not write another service's database. References across services use opaque IDs and versioned APIs/events. Combining catalogue and stock in one service keeps receiving, counting and stock changes atomic in the simple design.

13.2 Checkout consistency

Sales creates Pending sale with unique request ID > Inventory atomically validates stock/locks/quality and reserves FEFO quantities > Sales stores payment record > Inventory commits reservation and posts stock movements > Sales marks Completed and issues receipt. Stock commit and reservation status change shall be atomic. Completion shall require a confirmed commit.

Timeouts shall trigger status lookup by the same request ID. A reconciler shall finish committed sales or release uncommitted reservations. Reservations expire after a proposed five minutes; commit and expiry shall compete atomically so only one succeeds. If payment was externally accepted but reservation expires, owner/support shall resolve by a fresh reservation or recorded external refund; the system shall never silently claim a completed sale.

13.3 Return consistency and events

Sales shall prevent excess cumulative returns before requesting idempotent Inventory receipt/quarantine actions. Returns remain Pending until stock action and refund recording are confirmed. Refund-without-goods performs no stock addition. A reconciliation job shall recover partial failures; retries shall not create duplicate financial or stock records.

Owning services shall publish events using an outbox or equivalent durable mechanism. Consumers shall deduplicate event IDs. Reporting lag shall not block checkout; projections shall be replayable. Identity or Inventory failure shall block new checkout when authorization or availability cannot be established.

13.4 Interfaces

JSON REST APIs with versioning; authenticated internal requests; durable event broker. Browser printing for A4 and 80 mm. Manual scale reading and manual external-payment confirmation. Owner/admin write access shall be subscription checked; recovery workers may finalize previously authorized operations after expiry.

14. Non-functional requirements

These are proposed release targets. Verify on representative data: 100 active tenants, 50 concurrent users platform-wide, 10,000 SKUs and 50,000 batch positions per tenant, and 100,000 sales per tenant.

NFR-01 | Response time

95% of product/stock searches shall complete within 2 seconds and checkout within 3 seconds, excluding manual payment/printing. Twelve-month reports shall complete within 10 seconds or run asynchronously with a visible status. Reporting projections shall normally refresh within 60 seconds.

NFR-02 | Availability and failure isolation

Target 99.5% monthly availability excluding announced maintenance. Health alerts shall detect failures within five minutes. Reporting outages shall not stop sales. Outages shall show a clear error and preserve pending operation identifiers.

NFR-03 | Security and isolation

Use TLS 1.2 or later, salted adaptive password hashes, encrypted storage/backups, expiring tokens and login rate limits. Role and tenant checks shall apply to APIs, events, reports and exports. Disabled users shall lose protected access. Shield support access to tenant records shall require explicit role authorization and audit.

NFR-04 | Inventory and money integrity

Use decimal quantities/costs and monetary half-up rounding, never binary floating-point calculation. Stock changes/reservations shall be concurrency safe and idempotent. Immutable history and financial snapshots shall survive product/price changes. Failure tests shall prove no duplicate committed sale, refund or stock movement.

NFR-05 | Recovery

Daily encrypted backups retained 30 days; proposed RPO 24 hours and RTO four hours. Restore shall recover service data and replay/reconcile pending operations. Verify a restore before launch and quarterly. Unresolved transactions shall alert operations within five minutes.

NFR-06 | Usability and compatibility

Clear kg/item labels, visible stock and field errors, keyboard navigation and labeled controls. Trained cashier shall process five lines in 60 seconds excluding weighing/payment/printing. Verify current/previous Chrome and Edge at release, tablet checkout >=768 px and owner views >=360 px, with one validated 80 mm printer.

NFR-07 | Maintainability and observability

Versioned APIs, service documentation, automated critical-flow tests and deployment rollback shall be available. Central logs shall use tenant and correlation IDs without passwords/tokens/card credentials. Monitor checkout failures, stock discrepancies, expiry jobs and projection lag.

NFR-08 | Retention and privacy

Customer details are optional. Proposed transaction/audit retention is 12 months after subscription end, with owner export before approved deletion. Final retention and deletion policy shall be agreed before production. Deletion shall cover live data and documented backup expiry; no cross-tenant exposure is permitted.

15. Acceptance tests and requirements coverage

AT-01 | Access, setup and subscription

Create two tenants; verify all roles and cross-tenant denials. Expire a subscription, block new writes, permit owner export and complete already authorized reconciliation. Renew and restore access. Covers FR-01 to FR-07, FR-25; NFR-03.

AT-02 | Receipt and units

Run section 8 receiving example; verify exactly 18.000 kg added and LKR 400/kg cost. Partial receipt updates order outstanding quantity. Duplicate posting adds nothing. Reject negative tare, fractional EACH and excess receipt without authorization. Covers FR-08/09; NFR-04.

AT-03 | Quality, FEFO and availability

Create dated/undated batches, ready/unripe fruit, quarantine and past-use-by stock. Verify default FEFO, blocking rules and audited overrides. Two cashiers compete for the last 1.000 kg: only one reservation succeeds. Transfer 5.000 kg without changing total. Covers FR-10/11/14/16.

AT-04 | Loss and count

Clerk submits 0.750 kg spoilage; owner approves once. Check loss value and ledger. Start scoped count, block its stock movements, approve -0.200 kg once and unlock. Reject adjustment against reserved stock. Covers FR-12/13; NFR-04.

AT-05 | Checkout and receipt

Run the 1.250 kg mango sale example; verify net amount LKR 731.25 and change LKR 268.75. Confirm correct precision, price snapshot, batch allocation and receipt. Retry checkout, lose connection after Inventory commit, and reconcile to one completed sale. Covers FR-15 to FR-18; NFR-01/04/05.

AT-06 | Returns

Return 0.250 kg against the sale; calculate original proportional discount/tax refund. Verify quarantine, owner release, untraceable batch handling, refund-without-goods, cumulative limits and duplicate retry protection. Covers FR-19/20.

AT-07 | Reports and cash shifts

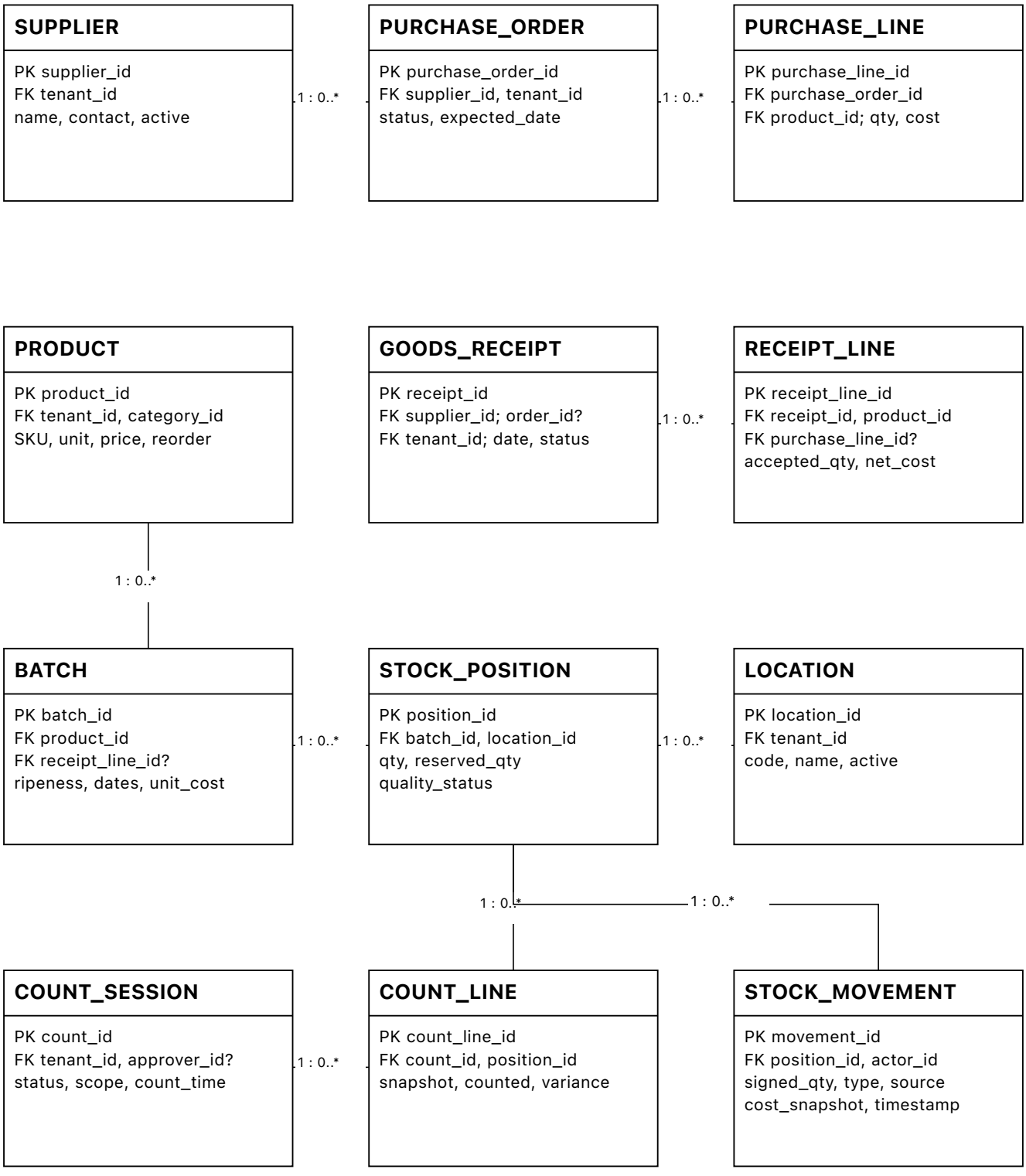
Reconcile net revenue, batch costs, spoilage and stock with ledgers. Show KG and EACH totals separately. Verify margin visibility, CSV export, projection timestamp, cash float/refunds/cash-in/out and closing variance. Walk-in sale works without customer data. Covers FR-21 to FR-25.

AT-08 | Operational release checks

Measure stated performance/load and reporting lag; validate restore, browser layouts, printing, usability, service failure alerts, disabled accounts and tenant isolation. Verify cancelled/rejected documents cause no stock change. Covers NFR-01 to NFR-08.

16. ER diagram - inventory domain

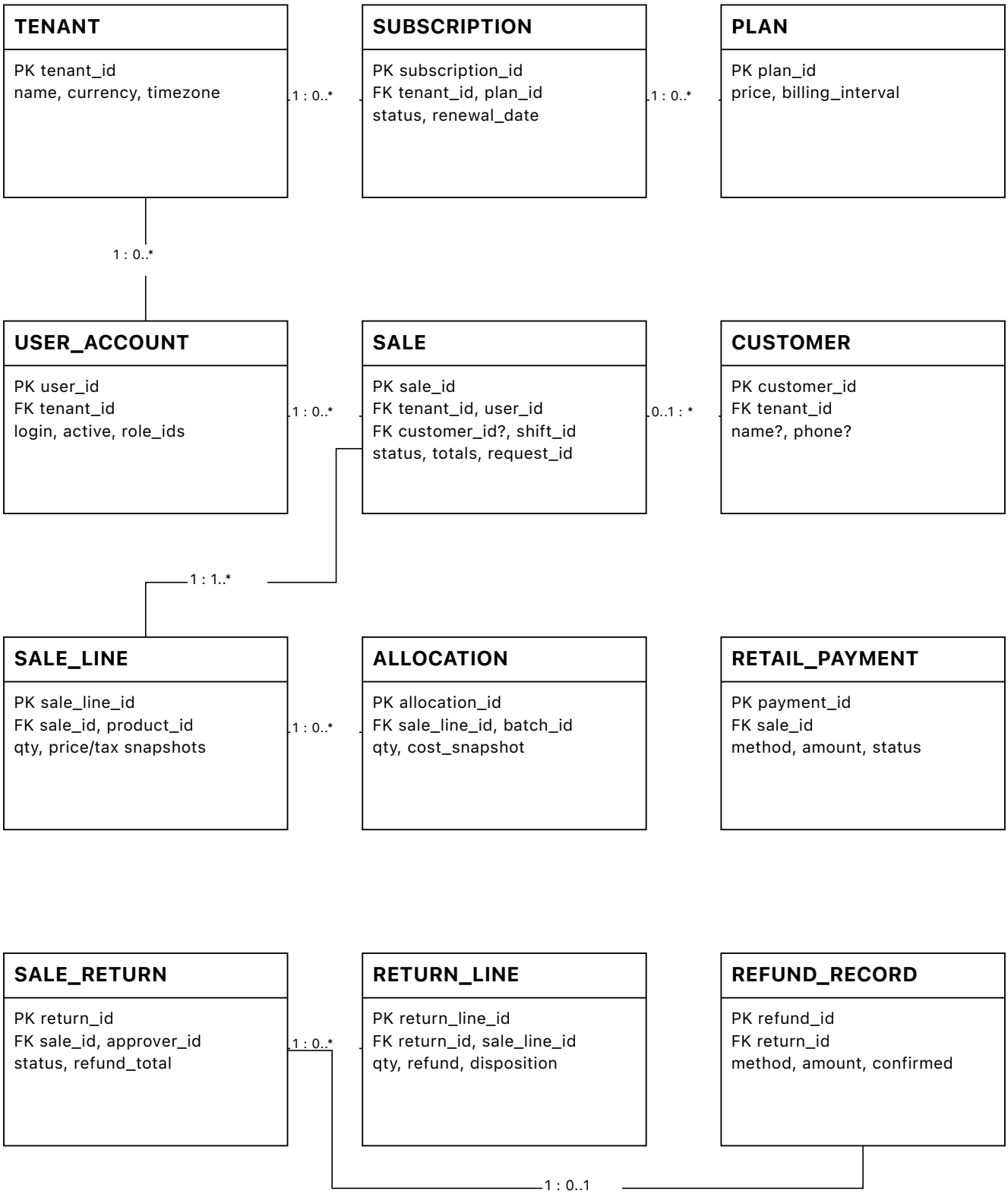
Logical entities | PK: primary key | FK: reference | ?: optional



Additional links: Receipt 1 -> 0..* ReceiptLine; Product 1 -> 0..* PurchaseLine / ReceiptLine.
ReceiptLine 1 -> 1 Batch after posting; Supplier 1 -> 0..* Receipt; Position 1 -> 0..* Movement.
CountLine belongs to one Position. Opening/return batches may have no ReceiptLine.
Each tenant-scoped FK must match tenant_id. Wastage/return documents link ledger sources.

17. ER diagram - sales and tenancy

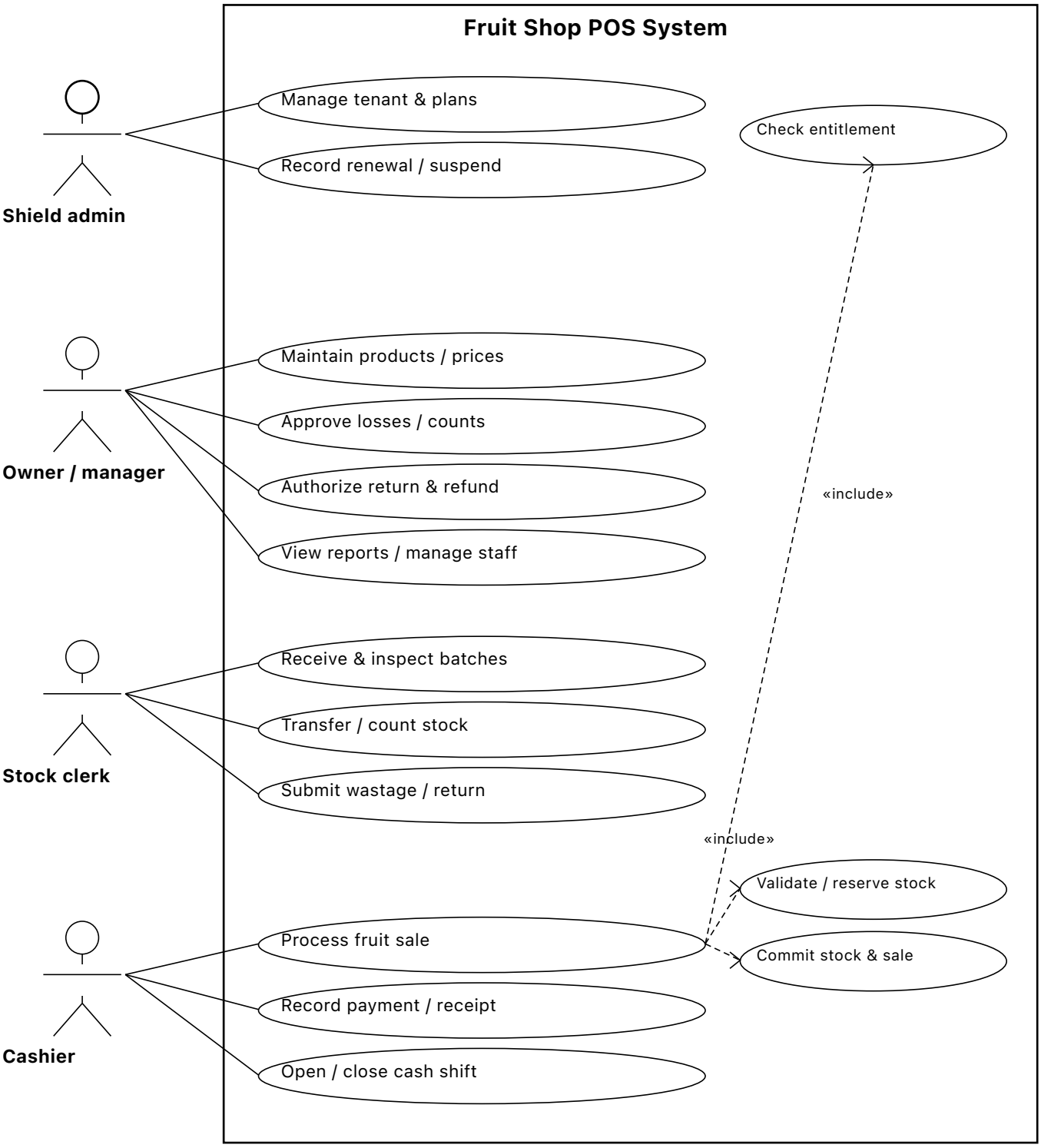
Logical entities | PK: primary key | FK: reference | ?: optional



Additional links: Sale 1 -> 1..* SaleLine and 0..1 RetailPayment (1 when Completed).
Sale 1 -> 0..* SaleReturn; SaleLine 1 -> 0..* ReturnLine; Return 1 -> 1..* ReturnLine.
Posted Return has 1 RefundRecord; Batch 1 -> 0..* Allocation; Product 1 -> 0..* SaleLine.
Cross-service IDs are logical references, not shared database foreign keys.

18. UML use case diagram

Actors, permissions and required subflows



Solid line: actor association. Dashed arrow: mandatory included use case.
Owner permissions also allow clerk and cashier actions (listed in the role matrix);
refund and stock approvals require the owner role. All protected writes check entitlement.

19. Operational exceptions and final decisions

19.1 Required exception behavior

Wrong received weight: owner reverses unused receipt and posts corrected receipt. If batch already sold or moved, use a reasoned stock/cost correction workflow preserving affected sale snapshots; do not edit history. Loss of connectivity: show Pending with request ID and retrieve outcome on reconnect. Scale mistake before payment: edit basket; after completion: owner-authorized return/correction.

Quality changes during checkout: Inventory shall revalidate quality at commit and reject if stock is now blocked; Sales shall surface a pending payment-resolution case if payment was already externally taken. A stock count cannot lock an actively reserved position; it shall wait for release or completion. Past-use-by stock remains unavailable regardless of cashier selection.

Spoiled customer return: refund according to owner approval, receive to quarantine if goods are physically returned, then post disposal; revenue refund and wastage shall remain separate. Owner stock adjustment shall never substitute for a refund. Subscription expiry shall not delete records or abandon pending transactions.

19.2 Deployment and operational ownership

Shield Technologies owns product development, hosting, backup, subscription administration and operational monitoring. Shop owners own product accuracy, entered weights, supplier details, quality assessments, pricing/tax settings and approvals. The software shall record these decisions and enforce configured controls; it shall not automatically determine fruit safety.

19.3 Decisions to approve before implementation

Confirm subscription prices/intervals, trial/grace policy (none assumed), stock clerk posting permissions, scale-entry precision, default saleable ripeness, retention policy, supported printer, production tenant/load targets, purchase cost treatment and alert windows. Current cost assumption: entered batch cost is net accepted line cost, excluding recoverable tax and without automatic freight allocation.

Baseline first-release exclusions remain: automatic scale integration, gateway charging, mixed-fruit bundles, multi-branch, offline mode, credit accounts and accounting settlement. Future changes require requirement IDs, impact review and approved SRS revision.

19.4 Approval

Product owner: Shield Technologies
Business reviewer: _____
Technical reviewer: _____
Approval date: _____
Document status: Draft v2.0 for review